17^{TH} Experiment

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.linear_model import LogisticRegression

from sklearn.datasets import make_classification


# Generate linearly separable data

X, y = make_classification(

    n_samples=100,

    n_features=2,

    n_redundant=0,

    n_informative=2,

    n_clusters_per_class=1,

    class_sep=2,

    random_state=42

)


# Train Logistic Regression

model = LogisticRegression()

model.fit(X, y)


# Create mesh grid

x_min, x_max = X[:,0].min()-1, X[:,0].max()+1

y_min, y_max = X[:,1].min()-1, X[:,1].max()+1


xx, yy = np.meshgrid(

    np.arange(x_min, x_max, 0.02),

    np.arange(y_min, y_max, 0.02)

)


# Predict
```

```
Z = model.predict(np.c_[xx.ravel(), yy.ravel()])
Z = Z.reshape(xx.shape)

# Plot
plt.figure(figsize=(6,4))
plt.contourf(xx, yy, Z, alpha=0.3)

plt.scatter(
    X[:,0], X[:,1],
    c=y,
    edgecolors='k'
)

plt.title("Linear Separability using Logistic Regression")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")

plt.show()
```